

Finite-Element Analysis Simulations of a Piezoelectric Bending Actuator Actuator



Tuning residual stress and layer thickness for a flat, crack-free active-guidewire tip

Parker Robles

Department of Mechanical Engineering, Stevens Institute of Technology
Advisor: Prof. Yong Shi · Industrial Mentor: Dr. Sundeep Mangla, M.D.

Introduction

An active thrombectomy guidewire steers to a clot with a multilayer piezoelectric tip that must sit **flat at rest** and bend only when the PZT is driven. Fabrication cool-down locks in residual stress that can pre-bend the tip.

Goal of Study

Use finite-element analysis to find the residual-stress and Si₃N₄-thickness that leave the tip flat after cool-down, and to remove the stresses that crack the brittle PZT.

Methods

Five thin-film layers (SiO₂ / TiO₂ / PZT / Au / Si₃N₄) are modeled in **HyperMesh** and solved in **Radioss**. The cool-down down (250 → 25 °C) is simulated; residual stress is imposed in imposed in Si₃N₄ as an effective expansion rate:

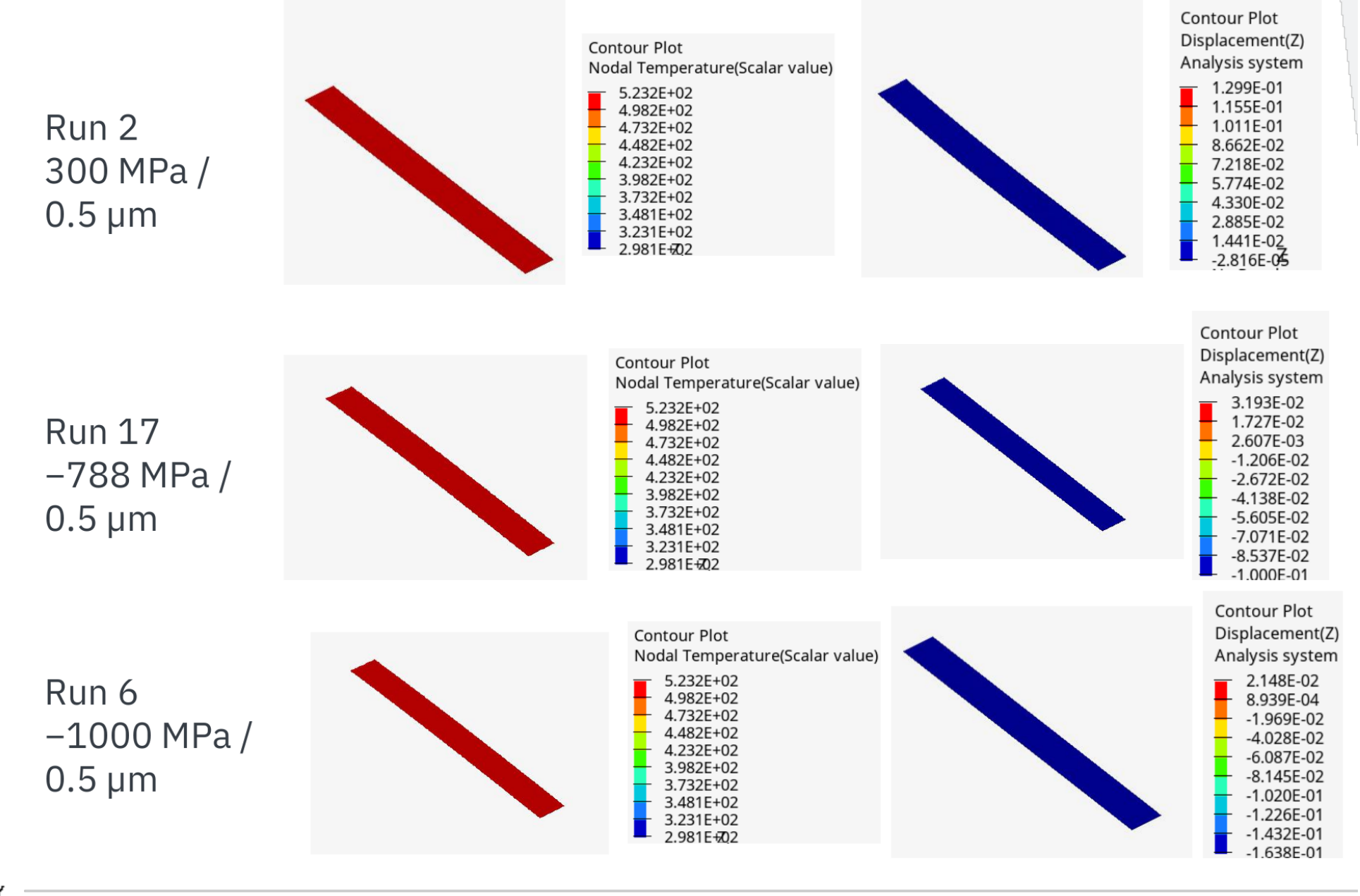
$$\sigma = E(\alpha_{\text{eff}} - \alpha_0) \Delta T$$

A Python loop (**flatten.py**) sweeps stress and thickness, reading the worst out-of-plane deflection (peak |Z|) until the tip is flat.

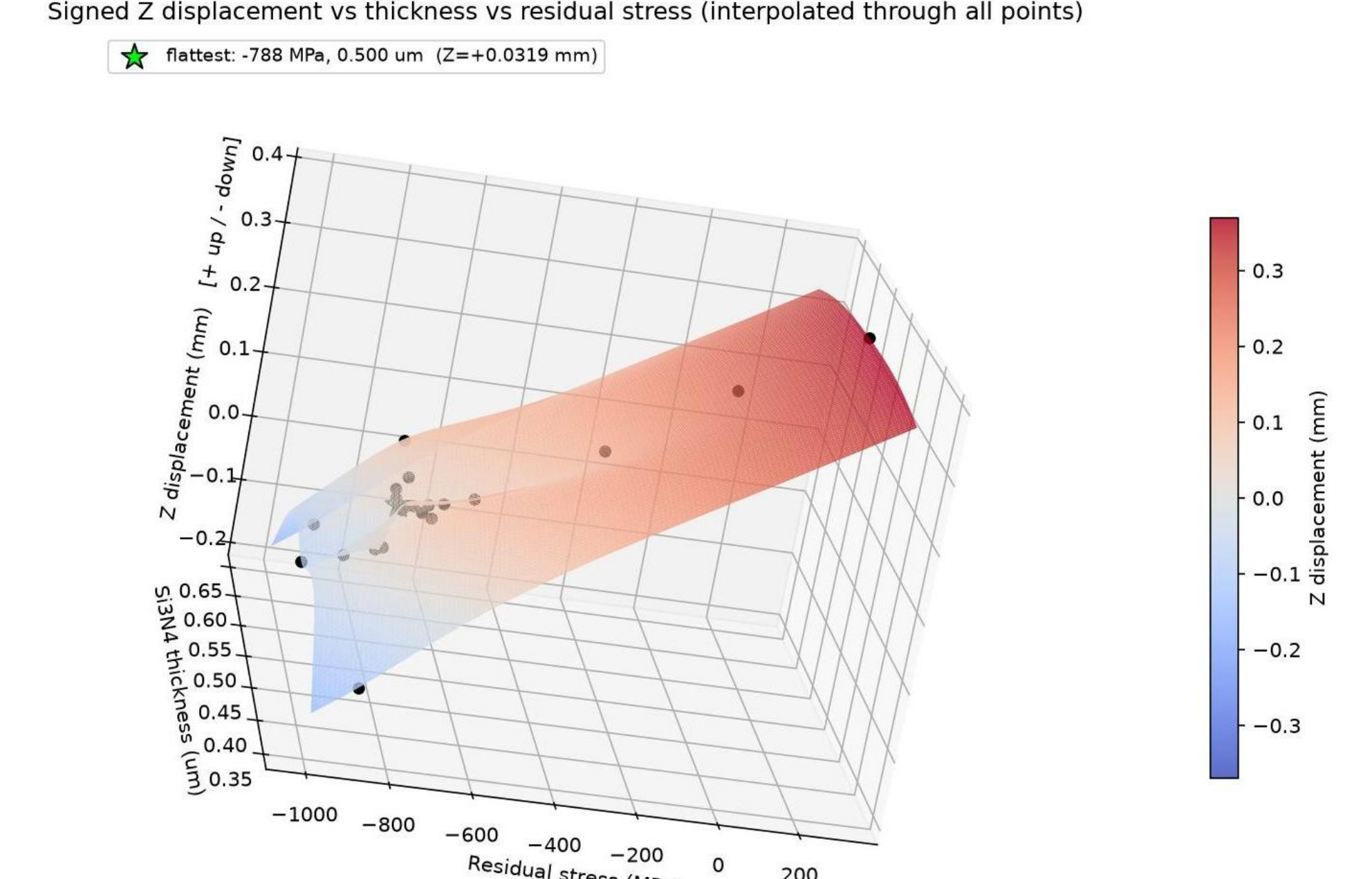
Material	E (MPa)	ν	ρ (Mg / m m³)	CTE (1/K)
SiO ₂	70,000	0.17	2.2×10 ⁻⁹	0.55×10 ⁻⁶
TiO ₂	150,000	0.27	4.2×10 ⁻⁹	9.0×10 ⁻⁶
PZT	75,000	0.31	7.5×10 ⁻⁹	4.0×10 ⁻⁶
Gold	79,000	0.42	1.93×10 ⁻⁸	14.2×10 ⁻⁶
Si ₃ N ₄	310,000	0.27	3.17×10 ⁻⁹	3.44×10 ⁻⁶

All layers ≈0.2 μm thick (TiO₂ 20.7 nm). Si₃N₄ carries the tuned residual stress; the TiO₂ interlayer buffers fabrication stress.

Results



FE Z-displacement contours — +300 MPa bends up, -788 MPa flat, -1000 MPa down.



Design space: signed Z-displacement vs. residual stress × thickness.

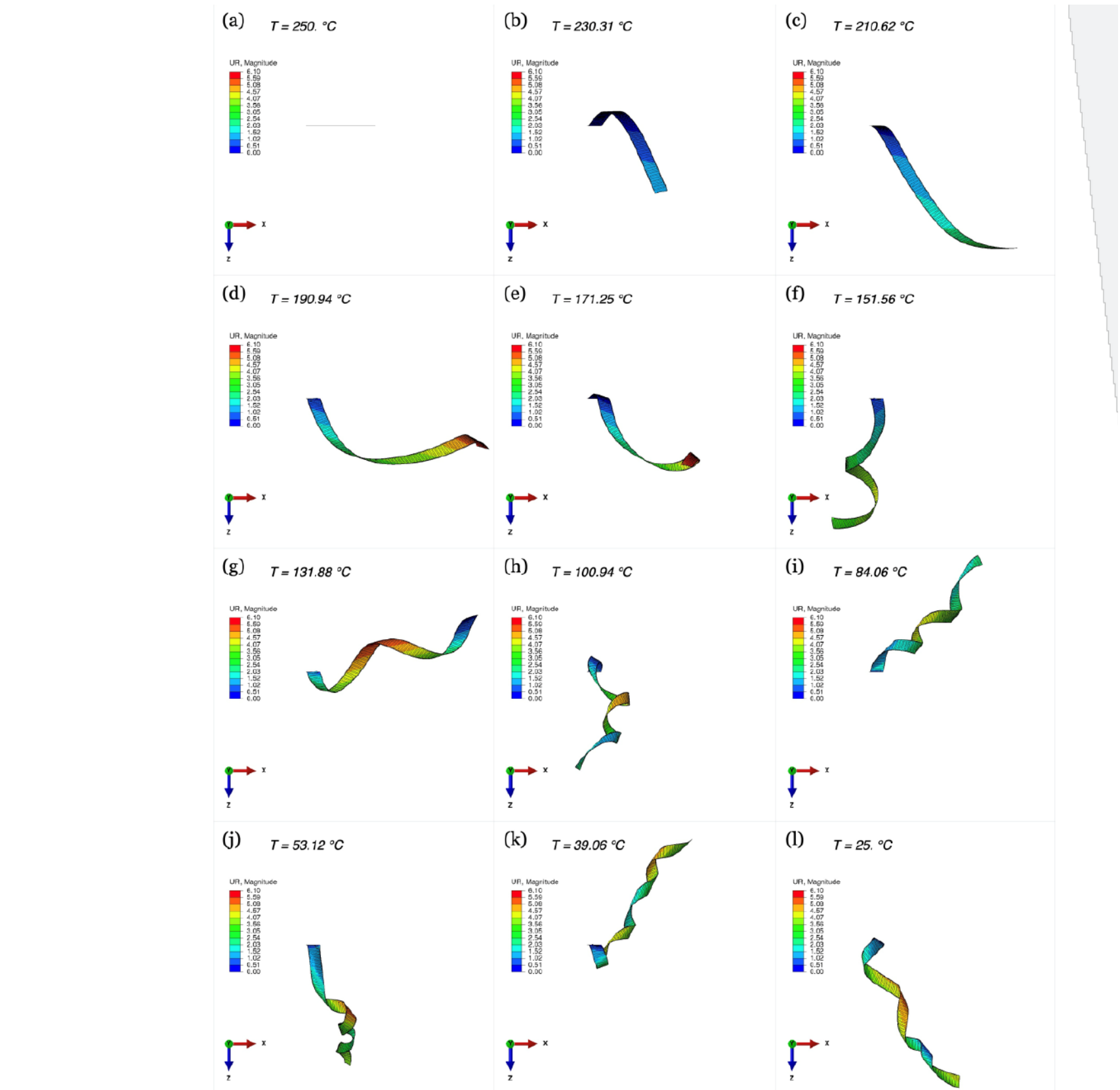
Key trends

- Residual stress sets bend direction — curl passes through **flat at -788 MPa** (peak |Z| = 0.032 mm).
- Twist (~0.06 mm) from the comb geometry sets the flatness floor.
- Thicker Si₃N₄ shifts the flat point toward less-negative stress.

Crack-free fabrication

A 20.7 nm TiO₂ interlayer absorbs the disk cool-down stress, protecting the PZT:

Layer	Stress	Status
SiO ₂	-14 MPa	Safe
TiO ₂	+ 342 MPa	Safe
PZT	≈ 0 MPa	No cracking



Cool-down deformation (250 → 25 °C): the actuator progressively curls as the layers contract unequally.

Conclusions & Future Prospects

- Residual stress of **-787.5 MPa with 0.5 μm Si₃N₄** gives the flattest tip (peak |Z| = 0.032 mm); the remaining deflection is a geometric twist floor.
- A TiO₂ interlayer eliminates PZT cracking during the fabrication cool-down.
- Future work:** combine the flat, crack-free design with PZT activation in a two-step cool → activate simulation, then fabricate the device.

References

- Powers WJ, et al. Guidelines for the Early Management of Acute Ischemic Stroke: 2019 Update. Stroke. 2019;50(12):e344–e418.
- Razban M, Dargahi J, Boulet B. Image-based Intraluminal Contact Force Monitoring in Robotic Vascular Navigation. arXiv:2012.10762, 2021.
- Kono K. Artificial Intelligence in Neuroendovascular Procedures. J. Neuroendovasc. Ther. 2025;19:2024-0107.
- Stoney GG. The tension of metallic films deposited by electrolysis. Proc. R. Soc. Lond. A. 1909;82(553):172–175.
- Altair HyperMesh & Radioss. Altair Engineering, 2024.

Acknowledgements

Thanks to Prof. Yong Shi (Technical Lead) and Dr. Sundeep Mangla, M.D. (Industrial Mentor), and the Active Guidewire team — M. Eraky, F. Bresa, B. Domingo, E. Emerton, J. Becker, and C. McGinley.